

Homework (Jalapeno)

Q1. The ancient Egyptians used $x^2 + 10x + 20$ to model the area of a rectangular field. What is the vertex form of this equation?

- (A) $(x + 5)^2 - 5$
 (B) $(x + 10)^2 - 80$
 (C) $(x + 5)^2 + 5$
 (D) $(x - 5)^2 + 5$
 (E) $(x - 10)^2 + 80$

Q2. Complete the square for $x^2 + 7x$ to find the vertex form of the equation.

- (A) $x^2 + 7x +$

(A) $x^2 - 7x -$

(A) $\text{left}(x + \text{right})^2$

(B) $\text{left}(x - \text{right})^2$

(C) $x^2 + 7x -$

(E) $(x - 1)^2 - 1$

Q5. The vertex form of a quadratic equation is $(x - 3)^2 + 2$. What is the standard form of this equation?

- (A) $x^2 - 6x + 11$
 (B) $x^2 + 6x + 11$
 (C) $x^2 - 6x + 9$
 (D) $x^2 + 6x + 9$
 (E) $x^2 - 6x + 7$

Q6. Complete the square for $x^2 - 9x$ to find the vertex form of the equation.

- (A) $x^2 - 9x +$

(A) $x^2 + 9x +$

(A) $\text{left}(x - \text{right})^2$

(B) $\text{left}(x + \text{right})^2$

(C) $x^2 - 9x -$

Q7. The ancient Chinese used $x^2 + 5x + 6$ to calculate the area of a triangular field. Convert this equation to vertex form.

- (A) $x^2 + 5x + 6$
 (B) $(x + 2.5)^2 - 3.75$
 (C) $(x + 2.5)^2 + 3.75$
 (D) $(x - 2.5)^2 - 3.75$
 (E) $(x - 2.5)^2 + 3.75$

Q8. The vertex form of a quadratic equation is $(x + 2)^2 - 3$. What is the standard form of this equation?

- (A) $x^2 + 4x + 1$
 (B) $x^2 + 4x - 1$
 (C) $x^2 - 4x + 1$
 (D) $x^2 - 4x - 1$
 (E) $x^2 + 4x + 4$

Open Ended Questions (FRQ)

Q9. Complete the square to rewrite $x^2 + 12x + 4$ in vertex form. Then, interpret the transformed equation.

Q10. The ancient Egyptians used the equation $x^2 - 8x + 15$ to model the height of a pyramid. Convert this equation to vertex form, then sketch the graph of the equation and identify the x-intercepts.

Chef's Tips

- Tip 1: Use the completing the square method to convert standard form to vertex form.
- Tip 2: Make sure to simplify the equation after completing the square.
- Tip 3: Always identify the vertex and x-intercepts of the equation after transforming it.